

QUANTS – GEOMETRY_22-9-2026

Directions for questions 1 to 3: Select the correct alternative from the given choices.

Find the area (in sq.cm) of an isosceles triangle whose base is 16 cm and each of the equal sides is 9 cm.

- ☐ a) $80\sqrt{17}$
- ☐ b) $24\sqrt{3}$
- ☐ c) $81\sqrt{16}$
- ☐ d) $8\sqrt{17}$

In $\triangle ABC$, $AB = AC = 9$ cm and $BC = 16$ cm. If M is the midpoint of BC , $\triangle ABM$ is right angled at M .

$$\therefore BM^2 + MA^2 = AB^2 \text{ or } MA^2 = AB^2 - BM^2 = 9^2 - 8^2 \text{ cm}^2$$

$$\Rightarrow MA = \sqrt{17} \text{ cm. The area of } \triangle ABC = \frac{1}{2} (BC) (AM) = \frac{1}{2} (16) (\sqrt{17}) \text{ cm}^2 = 8\sqrt{17} \text{ cm}^2$$

Choice (D)

The product of the lengths of two of the sides of a triangle equals half of the sum of the squares of the lengths of the same two sides. The triangle is

- ☐ a) equilateral.
- ☐ b) right angled but not isosceles.
- ☒ c) isosceles. (Correct Answer - ✓)
- ☐ d) scalene and acute angled.

Hide Correct Answer

Hide Solution

PREVIOUS

NEXT

Solution:

If the lengths of the two sides are a, b the given condition is $ab = (a^2 + b^2)/2$.
 $\therefore (a - b)^2 = 0$ or $a = b$. The triangle is isosceles.

Choice (C)

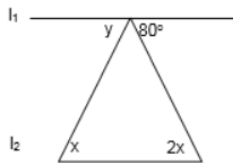
What is the ratio of the volumes of a cone, a cylinder and a hemisphere all having the same base area and the same height?

- ☐ a) 1 : 1 : 4
- ☒ b) 1 : 3 : 2 (Correct Answer - ✓)
- ☐ c) 1 : 2 : 3
- ☐ d) Cannot be determined

Solution:

The volume of a cone of base radius r and height h is $\frac{1}{3}\pi r^2 h$. The volume of a cylinder of base radius r and height h is $\pi r^2 h$. The volume of a hemisphere of radius r is $\frac{2}{3}\pi r^3$.
The required ratio is 1 : 3 : 2
Choice (B)

Directions for question 4: Type in your answer in the input box provided below the question.



In the figure, l_1 and l_2 are parallel lines. Find y . (in degrees)

Your Answer: 55 ✗

Correct Answer: 40 ✓

Solution:

As $l_1 \parallel l_2$, $y = x$ (Alternate Angles)
The third angle of the triangle (unlabelled)
 $= 180^\circ - 3x = 180^\circ - (80^\circ + x)$
 $\therefore 3x = 80^\circ + x \Rightarrow x = 40^\circ$
 $y = x = 40^\circ$ (alternate angles)

Ans : (40)

The radius of a circle circumscribing a triangle is 5 cm. Two sides of the triangle measure 6 cm and 8 cm and its area is 24 sq.cm. Find the third side.

- ☐ a) 10 cm
- ☐ b) 9 cm
- ☐ c) 6 cm

☒ d) 5 cm (Your answer - ✕)

Solution:

The area of the triangle, $A = \frac{1}{2} ab \sin C$. Here $a = 6$ cm, $b = 8$ cm and $A = 24$ cm².
 $\therefore \sin C = \frac{2A}{ab} = \frac{2(24)}{6(8)} = 1$, i.e. $C = 90^\circ$. The third side has to be 10. ($\because 6^2 + 8^2 = 10^2$)

Alternately,

$$\frac{abc}{4R} = \Delta$$
$$\Rightarrow \frac{6 \times 8 \times c}{4 \times 5} = 24 \Rightarrow C = 10$$

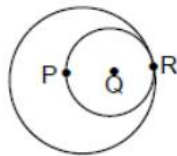
Choice (A)

The distance between the centres of two circles touching each other internally is 7 cm. What is the area common to both the circles, if the ratio of their radii is 2 : 1?

- ☐ a) 154 cm²
- ☐ b) 144 cm²
- ☐ c) 6164 cm²
- ☐ d) 1386 cm²

Solution:

The two circles are shown in the figure below.



As the radii are in the ratio $2 : 1$, the inner circle passes through the centre of the bigger circle, P. As $PQ = 7$ cm, the radius of the smaller circle is 7 cm. The area common to the two circles is the entire smaller circle, which has an area $\pi(7^2)$ cm^2 , or approximately, 154 m^2 .
Choice (A)

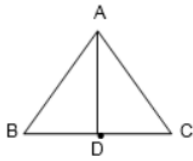
The areas of two similar triangles ABC and PQR are 36 sq.cm and 144 sq.cm respectively. Find the ratio of their corresponding sides.

- ☐ a) $1 : 2$
- ☐ b) $1 : 3$
- ☐ c) $1 : 4$
- ☐ d) $2 : 3$

Solution:

As the ratio of the areas of the similar triangles is $36:144$, i.e, $1 : 4$, the ratio of their corresponding sides is $1 : 2$.
Choice (A)

Directions for question 8: Type in your answer in the input box provided below the question.



In the figure, AD is the angle bisector of

$\angle$

BAC. AB =

$$\frac{3}{2}$$

AC =

$$\frac{3}{4}$$

BC. Find

$$\frac{BD}{AC}$$

Solution:

$$AB = \frac{3}{2} AC = \frac{3}{4} BC \Rightarrow AB : AC : BC = 3 : 2 : 4$$

As $AB : AC = 3 : 2$, $BD : DC = 3 : 2$
(Angle Bisector Theorem)

$$\therefore BD = \left(\frac{3}{3+2} \right) BC = \frac{3}{5} BC$$

$$\frac{BD}{AC} = \frac{3}{5} \quad \frac{BC}{AC} = \frac{3}{5} \quad \frac{4}{2} = \frac{6}{5} = 1.2$$

Ans : (1.2)

Directions for question 9: Type in your answer in the input box provided below the question

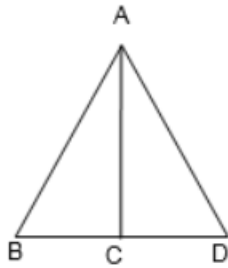
A circular lawn of area 38.5 sq. m is surrounded by a pavement of uniform width 0.7 m all around it. What is the area occupied by the pavement? (in sq. m)

Solution:

Let r be the radius of the lawn $\pi r^2 = 38.5 \text{ m}^2 \Rightarrow r^2 \times \frac{7}{22} (38.5) \text{ m}^2 = (3.5)^2 \text{ m}^2 \Rightarrow r = 3.5 \text{ m}$.

The width of the pavement is 0.7 m . The area of the pavement is $\pi(R^2 - r^2)$ where R is the radius of the bigger circle (the lawn plus pavement). Area = $\pi(R - r)(R + r) \approx$

$$\frac{22}{7} (0.7) (3.5 + 4.2) \text{ m}^2 = \frac{22}{7} (0.7) (7.7) \text{ m}^2 = 16.94 \text{ m}^2 \quad \text{Ans : (16.94)}$$



In the figure, $AB = AD$.

$\angle$

$ACB = 95^\circ +$

$\angle$

BAC .

$\angle$

$BAD = 150^\circ$. Find

$\angle$

ACB .

Solution:

An important part of the solution is redrawing the figure. The figure given in the question is only indicative.

As $\angle BAD = 150^\circ$ and $AB = AD$, all three angles of $\triangle ABD$ are known – $150^\circ, 15^\circ, 15^\circ$.

If $\angle BAC = x$, $\angle ACB = 95^\circ + x$.

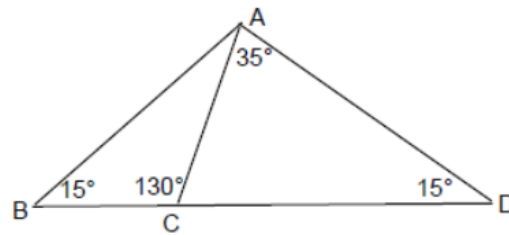
In $\triangle ABC$, $\angle A = x$, $\angle B = 15^\circ$, $\angle C = 95^\circ + x$

$$x + 15^\circ + 95^\circ + x = 180^\circ$$

$$\Rightarrow x = 35^\circ$$

$$\therefore \angle ACB = 95^\circ + x = 130^\circ$$

The figure should actually look as shown below



Choice (C)

The area and the perimeter of a rectangle are 48 sq. cm and 28 cm respectively. Find the ratio of its length and breadth.

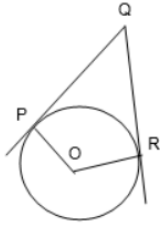
- ☐ a) 7 : 5
- ☐ b) 8 : 5
- ☐ c) 2 : 1
- ☐ d) 4 : 3

Solution:

Let the breadth and length of the rectangle be a cm, b cm $ab = 48$ and $a + b = 14$

$\therefore (a, b) = (6, 8)$ ($\because a < b$) the ratio of the length and breadth is 4 : 3.

Choice (D)



In the figure, QP and QR are tangents to the circle. If

$\angle$

$\angle PQR = 50^\circ$, find

$\angle$

POR.

- ☐ a) 130°
- ☐ b) 120°
- ☐ c) 110°

☒ d) 100° (Your answer - ✕)

Solution:

In quadrilateral OPQR, $\angle P = \angle R = 90^\circ$.

$\therefore \angle O + \angle Q = 180^\circ$

As $\angle Q = 50^\circ$, $\angle O = 130^\circ$.

Choice (A)

In a triangle of perimeter 66 cm, a circle of radius $5\frac{1}{3}$ cm is inscribed. What is the area of the triangle (in sq. cm)?

- ☐ a) 66
- ☐ b) 176
- ☐ c) 132
- ☐ d) 88

Solution:

The area of any triangle is rs , where r is the radius and s is the semi perimeter.

$$\therefore \text{Area} = \left(5\frac{1}{3}\right)(33) \text{ cm}^2 = 176 \text{ cm}^2$$

Choice (B)

In an isosceles trapezium PQRS with PQ parallel to RS, PT and QU are the perpendiculars drawn to the base RS from PQ and

$\angle$

QRU = 50° . Find

$\angle$

PST.

- ☐ a) 50°
- ☐ b) 60°
- ☐ c) 65°
- ☐ d) 70°

Solution:

In any isosceles trapezium PQRS, with PQ parallel to RS, $\angle P = \angle Q$ and $\angle R = \angle S$.
As $\angle R = 50^\circ$, $\angle S$ is also 50° .

Choice (A)

In the previous question, find

$\angle$

PQR

- ☐ a) 130°
- ☐ b) 165°
- ☐ c) 150°
- ☐ d) 125°

Solution:

As $\angle Q$ is the supplement of $\angle R$ and $\angle R = 50^\circ$, $\angle Q = 130^\circ$

Choice (A)

What is the area (in sq.cm) of a rhombus whose diagonals measure 46 cm and 54 cm?

- ☐ a) 1242
- ☐ b) 2484
- ☐ c) 224
- ☐ d) 1123

Solution:

The area of a rhombus is half the product of its diagonals, i.e. $\frac{1}{2} (46) (54) \text{ cm}^2 = \frac{2484}{2} \text{ cm}^2 = 1242 \text{ cm}^2$

Choice (A)

Directions for questions 17 and 18: Type in your answer in the input box provided below the question

Find the distance between the parallel sides of a trapezium whose area is 110 sq.cm and the parallel sides measure 8 cm and 14 cm. (in cm)

Solution:

The area of a trapezium is $\left(\frac{a+b}{2}\right)h$, where a, b are the lengths of the parallel sides and h is the perpendicular distance between them.

$$\therefore h = \frac{2A}{a+b} = \frac{2(110)}{8+14} \text{ cm} = 10 \text{ cm}.$$

Ans : (10)

Directions for questions 17 and 18: Type in your answer in the input box provided below the question

If each interior angle of a regular polygon is 150° , find the number of diagonals of the polygon.

Solution:

Each exterior angle is 30° . The sum of all the exterior angles of any convex polygon (not necessarily regular) is 360° .

$\therefore$ There are 12 exterior angles, i.e, 12 interior angles. For an n-g on the number of diagonals is $n(n - 3)/2$. For a 12 – sided (a duo decagon), the number of diagonals is $12(9)/2$, i.e, 54.

Ans : (54)

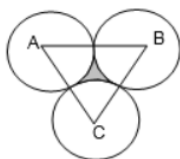
What is the ratio of the areas of two sectors cut from a circle if the ratio of their central angles is 6 : 5?

- ☐ a) 12 : 25
- ☐ b) 36 : 25
- ☐ c) 6 : 5
- ☐ d) 5 : 6

Solution:

If the ratio of the angles of the sectors is 6 : 5, the ratio of their areas (as well as the ratio of the arc lengths) is 6 : 5.
Choice (C)

What is the area of the shaded region in the following figure, if the radius of each circle 14 units and A, B, C are the centres of the circles (in sq. units)?



- ☐ a) $49(\sqrt{3} - 1)$
- ☐ b) $7(7\sqrt{3} - 2)$
- ☐ c) $28(7\sqrt{3} - 11)$
- ☐ d) $49(\sqrt{3} + 1)$

Solution:

The shaded region is the area of the equilateral triangle ABC minus the area of the 3 sectors (or half a circle of radius 7.)

$$\begin{aligned} \text{Area} &= \frac{\sqrt{3}}{4} (28)^2 - \frac{\pi(14^2)}{2} \approx \sqrt{3} (14^2) - \frac{22}{7} \frac{(14^2)}{2} = 14^2 \left[\sqrt{3} - \frac{11}{7} \right] \\ &= 14(2) (7\sqrt{3} - 11) = 28 (7\sqrt{3} - 11) \end{aligned}$$

Choice (C)